A prototype direct manipulation music tool based on Lerdahl's Pitch Spaces

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Abstract

Lerdahl (1988) suggests a model of 'Pitch Spaces' that treats pitches, chords and regions in a single framework. In this paper we will both summarise Lerdahl's arguments for the relative strengths of the model over previous cognitive models of harmonic knowledge, and we will present an extension to the model facilitating the design and implementation of a computer-based tool for harmonic analysis and composition for musical novices. We describe the current prototype of our computer music tool, and present plans for further development and evaluation of the tool with musical novices.

1 Introduction

Lerdahl (1988) suggests a model of 'Pitch Spaces' that treats pitches, chords and regions in a single framework. In this paper we will both summarise Lerdahl's arguments for the relative strengths of the model over previous cognitive models of harmonic knowledge, and we will present an extension to the model facilitating the design and implementation of a computer-based tool for harmonic analysis and composition for musical novices. Lerdahl's model, and the work presented in this paper are based around the Western Tonal Music system of harmony.

1.1 Motivation for another pitch space framework

Lerdahl's pitch space framework is not based on a SYMMETRICAL topological model - unlike those such as Longuet-Higgins (1962) and Shepard (1982). Lerdahl's states that a weakness of topological models is an over emphasis on symmetry - therefore misrepresenting the non-symmetrical aspects of the diatonic system. The pitch space framework is constructed with to be able to model the same asymmetry as the chords and keys it is modelling. Another strength of Lerdahl's model is that it is able to model more than one level of pitch representation in a single framework - previous systems have modelled either pitch classes (such as Balzano 1982 or Shepard 1982), or tonal regions (for example Weber 1824 and Schoenberg 1911/1978). Although one existing tonal framework does model multiple levels of pitch description - Longuet-Higgins (1962) - that system is applied to tonal regions (derived from pitch classes) and cannot be used to describe pitch proximity.

Previous work cited by Lerdahl as influential on the development of Pitch Spaces providing descriptions of pitch classes, chord spaces and tonal regions included Krumhansl (1979 & 1983), Krumhansl, Bharucha & Kessler (1982) and Krumhansl & Kessler (1982).

2 Description of Lerdahl's pitch space framework

Suggested by the chromatic, diatonic and triadic 'overlearning' ideas of Deutsch & Feroe (1984) and Deutsch (1982). Lerdahl's framework is a hierarchy of five spaces. The hierarchy is such that each level is made up of a subset of those pitch classes from the level immediately below. The five spaces are shown in Table 1.

Table 1: Names of pitch spaces

Level	Name
a	Octave space
b	Open fifth space

c	Triadic space
d	Diatonic space
e	Chromatic space

Important points Lerdahl makes about the spaces are as follows:

- except for the chromatic space, the spaces describe the asymmetric patterns appropriate for diatonic music
- the diatonic space is DIRECTLY represented in the framework (unlike the symmetrical frameworks mentioned earlier)
- the pitch space framework allows unified treatment of pitch class, chord and regional proximity

Lerdahl uses the Roman-numeral notation of chord / region - where we shall represent the region numeral in parentheses. For example. I/(I) is the pitch space for the tonic chord (say C major), in the region of the tonic (the C major diatonic region).

The choice of the tonic as C is arbitrary, and could be any other pitch class. The numeric forms for the pitch space I/(I) is shown in Table 2.

Table 2: Pitch space for I/(I)

Space	**Pitch Class**											
a	0											
b	0							7				
c	0				4			7				
d	0		2		4	5		7		9		b
e	0	1	2	3	4	5	6	7	8	9	a	b
e.d.	**0**	**4**	**3**	**4**	**2**	**3**	**4**	**1**	**4**	**3**	**4**	**3**

The final row, ED, is the 'embedding distance' - this is a measure of how far from the octave space a given pitch class is for a given pitch space. This distance shifts for a given chord and region. The shallower the embedding (such as 0 and 7 in I/(I)) the more important the pitch class harmonically for a given space.

Lerdahl explains this vertical embedding distance measure in terms of 'skip' and 'step':
"In traditional usage a STEP occurs between adjacent members of the chromatic or diatonic scales (a chromatic or diatonic step), and an ARPEGGIATION takes place between adjacent members of a triad. It is more illuminating, however, to thing of an arpeggiation as stepwise motion in triadic space [space c]. A leap of two octaves, on the other hand, is a skip in octave space [space a]. In sum, a step is adjacent motion along any level of the hierarchy, and a skip is non-adjacent motion -- two or more steps -- along any level."
(Lerdahl, 1988, pp. 321-322)

Therefore, using Lerdahl's definition of step and skip, the proximity of two pitches in a given pitch space (e.g. I/(I)) can be measured as a 'step distance' by the number of steps left or right at a given level to get from one pitch to another - e.g. in I/(I) from p0 to p4 is one step in triadic space, two steps in diatonic space and four steps in chromatic space. Chord proximity can be calculated using two factors: the diatonic circle of fifths and the number of common tones between the two chords. Lerdahl describes how each of these factors can be modelled via his pitch spaces. He presents the 'chord circle rule', defined as instruction to "move the pcs [pitch classes] at levels a-c four diatonic steps to the right of left (mod 12) on level d" (p. 322). Thus there is no change to the diatonic or chromatic spaces when modelling the chord circle. The circle of fifths
[pc0 (I) - pc7 (V) - pc2 (ii) - pc9 (vi) - pc4 (iii) - pcb)IV) - pc0 (I) and so on] appears as a sequence of pitch spaces when the chord circle rule is successively applied as shown from I/(V) to I/(ii) in tables 3 and 4.

Table 3: Pitch space for V/(I)

Space	Pitch Class											
a								7				
b			2					7				
c	0		2					7				b
d	0		2		4	5		7		9		b
e	0	1	2	3	4	5	6	7	8	9	a	b

Table 4: Pitch space for ii/(I)

Space	Pitch Class											
a			2									
b			2							9		
c			2			5				9		
d	0		2		4	5		7		9		b
e	0	1	2	3	4	5	6	7	8	9	a	b

Lerdahl goes on to define a common tone distance between any two chords based on the number of distinct PCs and the shortest number of steps on the circle of fifths. He goes further, and defines a measure of chord proximity across regions using a rule that gives a chromatic circle of fifths, and a measure of region proximity and how seventh and minor chords can be modelled in the framework.

The hierarchical and numerical nature of the pitch spaces, and the simplicity of the measurement of pitch, chord and region proximity suggest the use of this framework for computational modelling. The pitch space formalism has strong explanatory power, and as Lerdahl goes on to discuss, appears to correlate with experimental results investigating pitch class stability (see Krumhansl 1979, and Krumhansl & Shepard 1979), multi-dimensional scaling of diatonic triads (Krumhansl, Bharucha & Kessler, 1982) and abstract region spaces (Krumhansl & Kessler, 1982).

3 Our extended pitch space model

The contribution we make to Lerdahl's pitch spaces is to make the model a circular model, rather than a tabular, linear arrangement. This new model is a two-dimensional model in the form of concentric circles - with chromatic space as the innermost circle, and octave space as the outer one. The result is that steps and skips are seen as motions around the circles to move from one space to another, and rules such as the circle-space rule are seen as rotations of the circles themselves.

An example of the model for I/(I) is presented in Figure 1[1].

[1] See Figures LAST-1 and LAST-2 for clearler versions of this screen in numeric (0..b) and note-letter (C..B) notations.

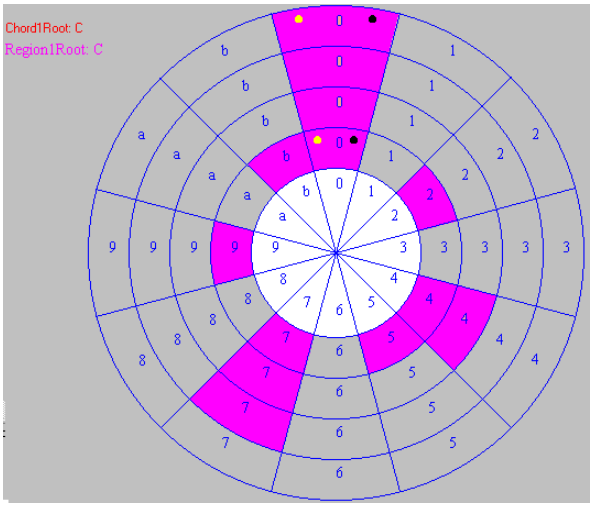


Figure 1: I/(**I**)

The modulus arithmetic required for calculating pitch classes becomes natural and intuitive with a circular organisation — if it is necessary to go beyond pitch class zero (e.g. C) or pitch class 11 (e.g. B) one just continues to count around the circle.

This straightforward extension to Lerdahl's original work makes the model much more suitable for computer-based implementation as a tool for music analysis and generation by musical novices.

4 The current prototype

A direct manipulation (see Shneiderman, 1982) prototype of our interactive, circular model of Lerdahl's pitch spaces has been developed, although it is still in its early stages.

Figures 2 and 3 show the current prototype running with I/(**I**) and V/(**I**) (shown as chords Cmaj and Gmaj in region Cmaj).

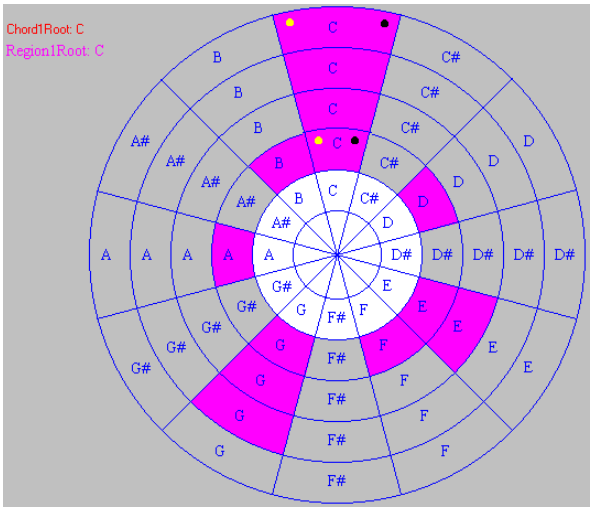


Figure 2: I/(**I**) as C/(C)

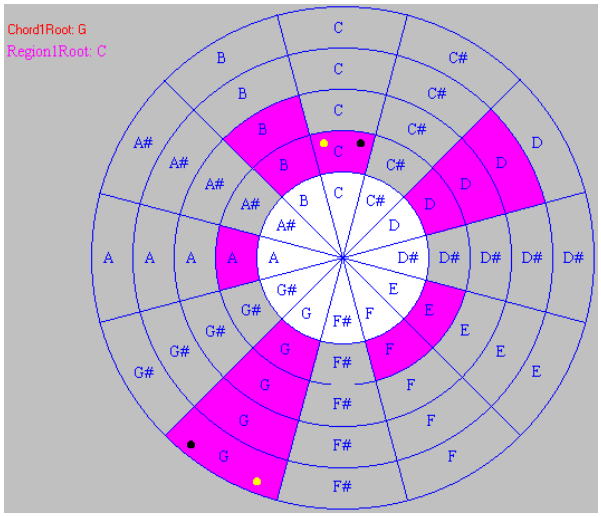


Figure 3: V/(**I**) as G/(C)

As you can see from the difference between Figure 1 and Figure 2 (and full versions in the web browser in Figures 9 and 10) the prototype allows the user to choose whether to see pitch class numbers (0..b or 0..11) or the note letters assuming pitch class 0 is C.

The user currently has simply facilities such as stepping (rotating) the major chord circles (levels a, b and c), and/or stepping the diatonic region circle (level d). The chromatic circle is left unchanged[2].

5 Proposed experimental evaluation

We are in the process of extending our prototype for use in experiments with novice musicians for the support of simple harmonic analysis and composition tasks. If Lerdahl's claims are correct, and the asymmetry of his model provides a good cognitive fit with human harmonic problem solving, we expect to find positive results from our trials, in comparisons with alternatives such as Holland's (1989) "Harmony Space" interactive computer tool.

In the next few sections we shall present some examples of the kinds of questions musical novices might be asked, and how they might be able to answer them using the tool.

5.1 Chords in a region

The musical novice might be presented with the system set up with Cmaj chord in the region Cmaj (as in Figure 2). They could be asked the following question:

In addition to Cmaj what are the other major chords likely to sound nice in this region?

A likely way to try to answer this question would be to keep the chromatic and diatonic circles (spaces 'd' and 'e') as they are, and to rotate the chord circles ('a', 'b' and 'c') to find chords which have common pitch classes with the current diatonic region (i.e. Fmaj and Gmaj).

Rotating clockwise from Cmaj, the student would first come to C#/(**C**) (see Figure 4). Clearly this chord does not share all notes with the region Cmaj, since neither C# nor G# as in Cmaj. Continuing to step the major chord shape (rotate the outer three circles) around the chromatic or diatonic circles the student would first come to chord Fmaj (see Figure 5) then Gmaj (see Figure 3) and find that they, in addition to Cmaj, are the only three major chords that have all notes common with the Cmaj region.

[2] Although it might make sense, and be more consistent, to allow the chromatic circle to be rotated to indicate when region **I** is associated with a note other than C.

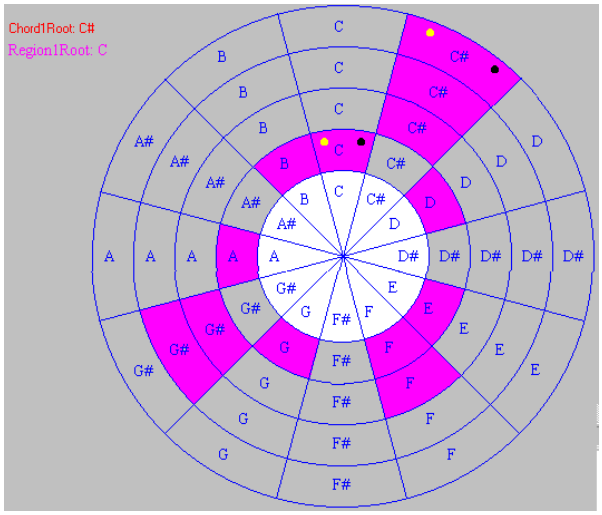


Figure 4: C#/(C)

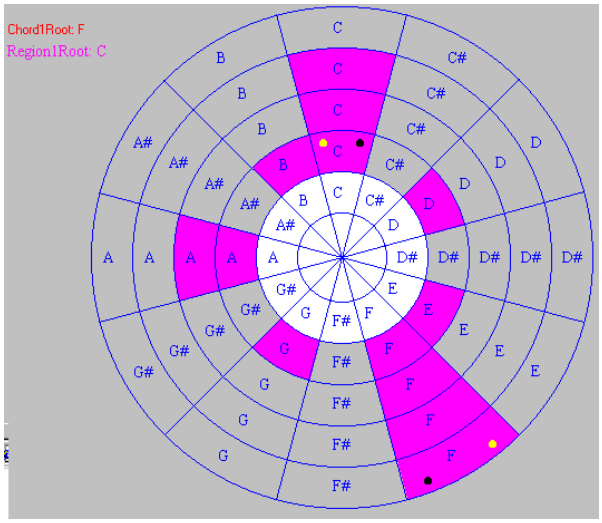


Figure 5: IV/(**I**) as F/(C)

5.2 Regions for which a chord will fit

Another question we might ask a student might be given a chord, in which regions the chord would share all three notes. For example if, once again, the musical novice is presented with the system set up with Cmaj chord in the region Cmaj they could be asked the following question:

In addition to the region Cmaj what are the other regions in which the chord Cmaj is likely to sound nice?

Rotating the diatonic region circle clockwise from Cmaj, the student would first come to C#maj (see Figure 4). Clearly the chord Cmaj only shares one note with the region C#maj (note C, see Figure 6).

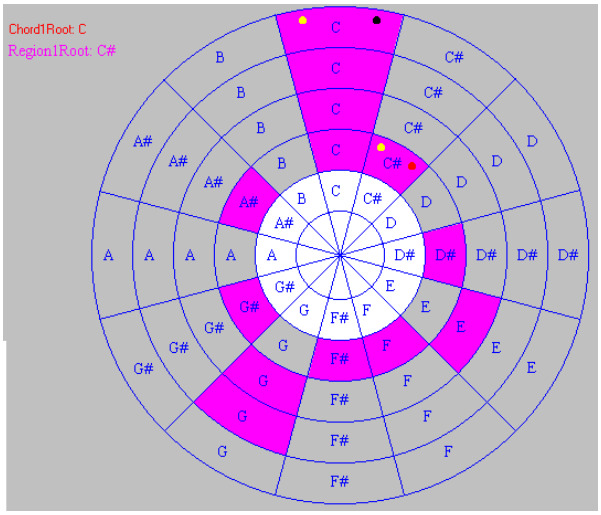


Figure 6: C/(C#)

Continuing to step the region around this way the student will come across the 2 diatonic regions in which chord Cmaj does shall all notes — region Fmaj (see Figure 7) and region Gmaj (see Figure 8).

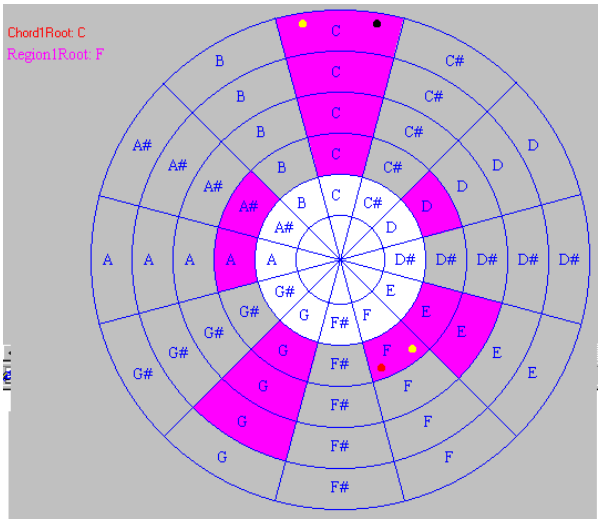


Figure 7: C/(F)

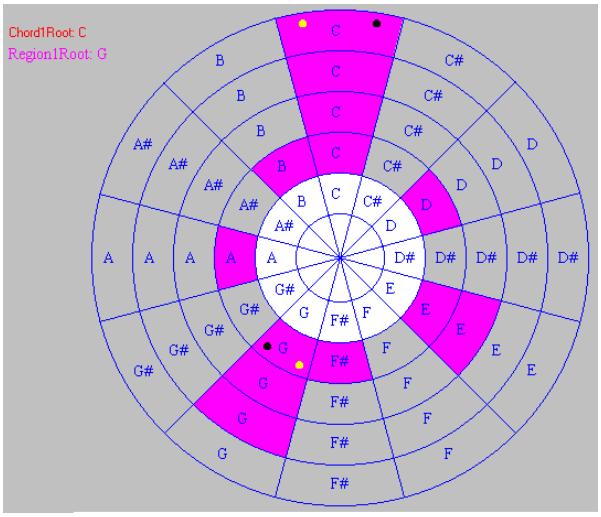


Figure 8: C/(G)

5.3 Regions that only different by one pitch class

In a similar fashion we could ask the student to derive the cycle of fifths:

Which are the closest regions to Cmaj — i.e. which regions shall all but one pitch class.

6 Conclusions & Further work

Initial, informal experiments with musical novices have been encouraging. Clearly the extended circular model maintains the features and strengths of Lerdahl's original, tabular pitch space model. Once the prototype implementation is complete stand along and comparative experiments as suggested above will be conducted. The fact that the model and computer program represent the asymmetry of the diatonic system may be important to help students move more easily from theory to practice on physical instruments where such asymmetry is unavoidable.

Extensions to the prototype being worked on include the following:

- A unified way to enable users to step and skip at any level
- The development of the tool to allow two (or more) different chords and/or regions to be visually compared (so to provide a graphical representation of Lerdahl's metrics)
- (related to above) the plotting of melodies on a circle, whereby users can choose a new set of circles each time a chord or region changes
- at present it is not possible to change the default that pitch class 0 is associated with C — although a trivial change it will enable more sophisticated and region-independent testing while still allowing students to refer to note letters if they wish

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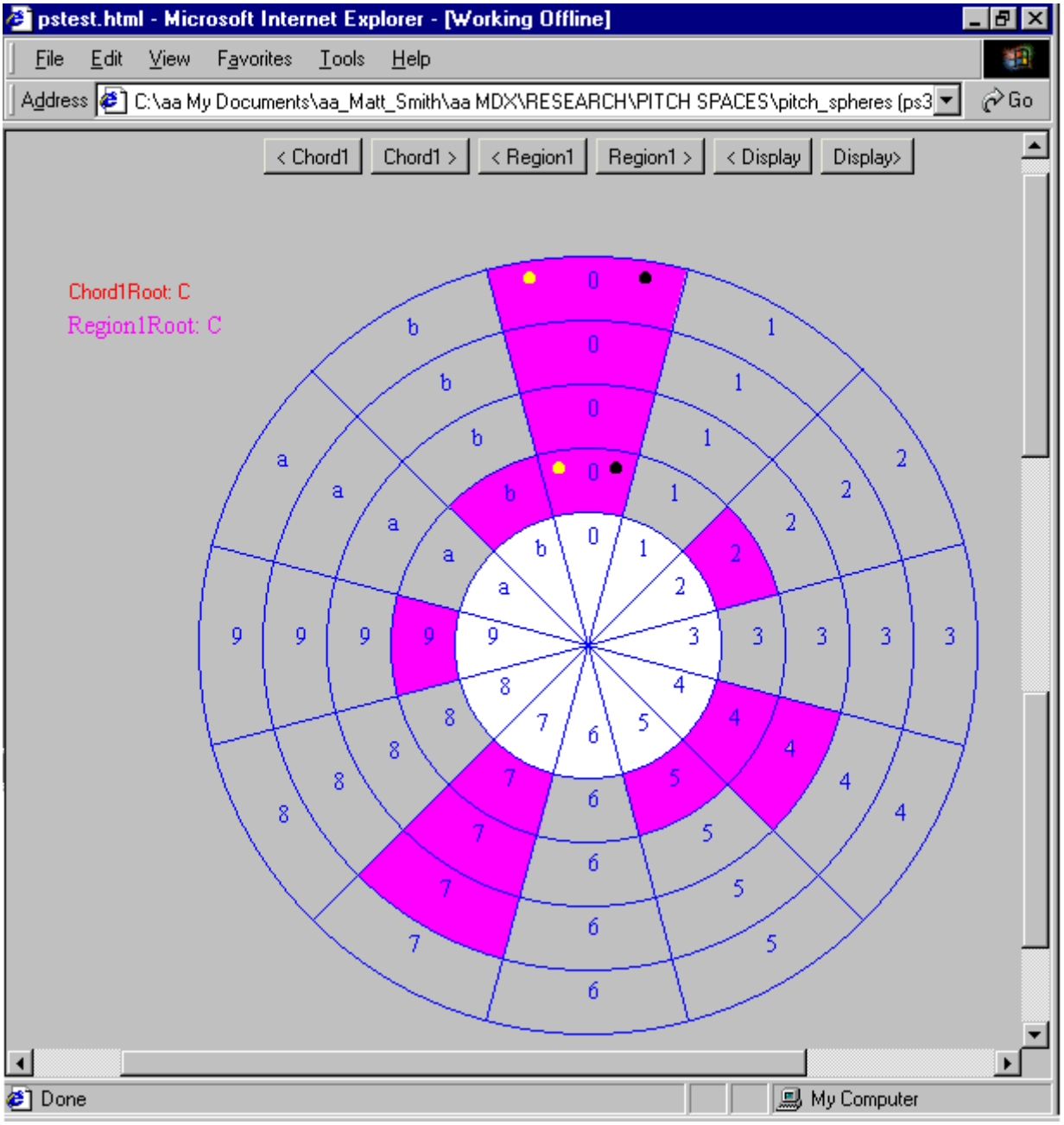


Figure 9: Prototype tool in web browser window, numeric notation mode — I/(**I**)

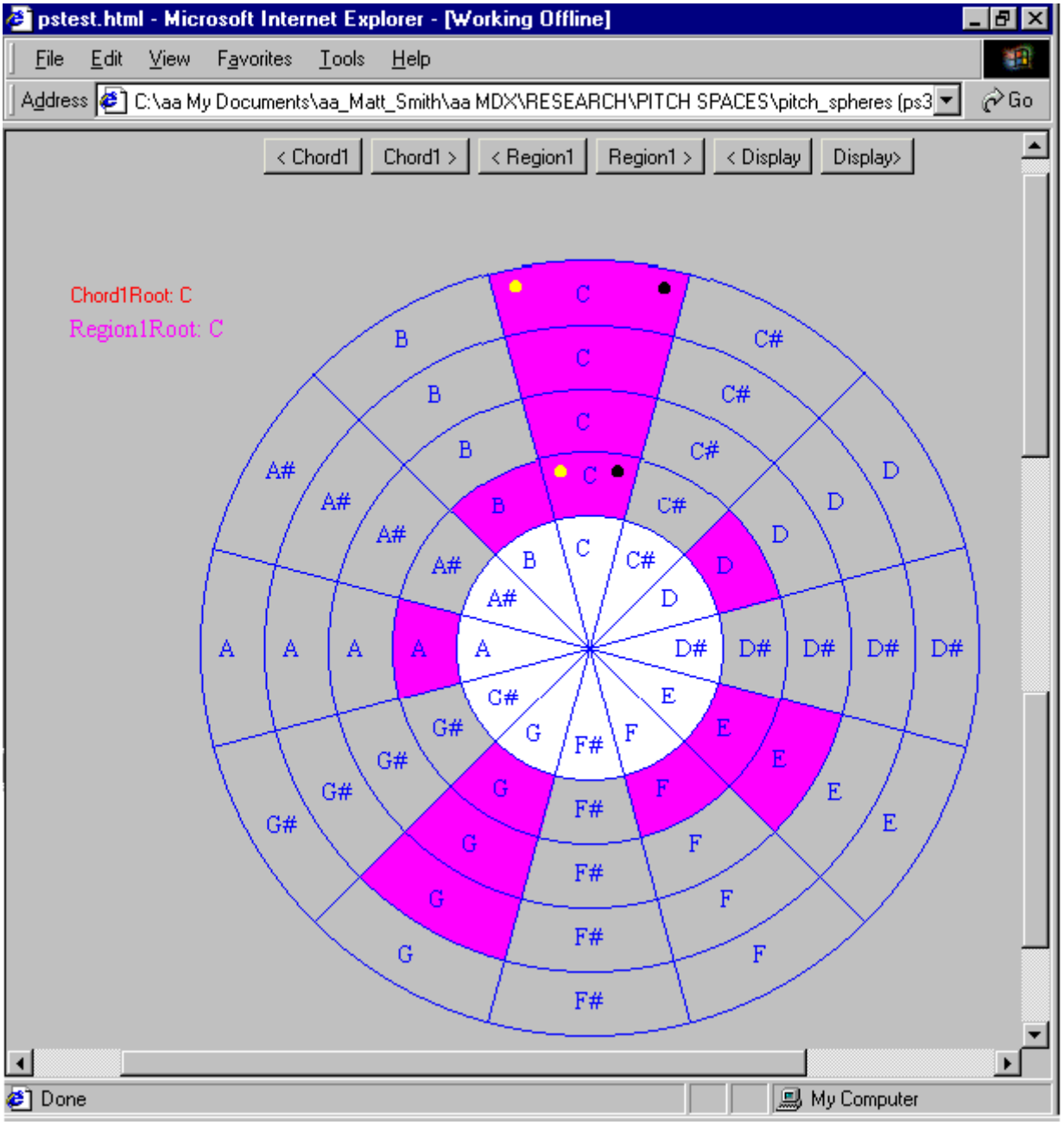


Figure 10: Prototype tool in note letter mode — I/(**I**) as C/(**C**)